

Efficiency of model-free screening under shared volatility

mfscreen2 simulation

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1 Motivation

The previous simulations compare screening methods in standard settings where the null predictors are close to the assumptions behind Pearson and Spearman correlation p-values. In those cases, all methods can have good FPR calibration.

Here the goal is to create a more diagnostic simulation for the efficiency of the reciprocal model-free statistic. The question is:

Can a method retain the true active variables while avoiding false positives caused by a common volatility factor?

This is relevant for high-dimensional screening because some variables can be selected not because they have a mean association with the response, but because they share observation-level volatility, outliers, or scale heterogeneity with the response. A good screening statistic should not confuse this volatility artifact with a useful marginal signal.

2 Competing Methods

We compare the same three marginal screening rules.

2.1 Model-free reciprocal statistic

For predictor X_j , let

$$z_{ij} = X_{ij} - \bar{X}_j, \quad u_i = Y_i - \bar{Y}.$$

The reciprocal statistic is

$$D_j^2 = \frac{\sum_i z_{ij}^2 u_i^2}{(\sum_i z_{ij} u_i)^2} - \frac{2 \sum_i z_{ij}^3 u_i}{(\sum_i z_{ij} u_i) (\sum_i z_{ij}^2)} + \frac{\sum_i z_{ij}^4}{(\sum_i z_{ij}^2)^2}.$$

Small values indicate stronger marginal evidence. At nominal level q , the selection threshold is

$$D_j^2 \leq \frac{1}{\{\Phi^{-1}(1 - q/2)\}^2}.$$

The denominator uses the empirical second moment of $z_{ij} u_i$, which makes the statistic sensitive to heteroskedasticity in a way that ordinary correlation p-values are not.

2.2 Pearson SIS

Pearson SIS computes the ordinary Pearson correlation r_j between X_j and Y , converts it to a two-sided p-value using the usual t -test, and selects if

$$p_j \leq q.$$

This is efficient under homoskedastic linear Gaussian assumptions, but the p-value is not designed for common volatility or strong scale heterogeneity.

2.3 Spearman SIS

Spearman SIS applies the same p-value rule to ranks. It is less sensitive to extreme values than Pearson SIS, but the rank p-value is still a simple correlation calibration and can be affected by common volatility.

3 Simulation Design

We use fixed

$$n = 1000, \quad p = 1000,$$

with active set

$$S = \{1, 2, 3, 4, 5\}.$$

The number of Monte Carlo replications is

$$B = 150.$$

The q grid is

$$0.001, 0.005, 0.01, 0.025, 0.05, 0.1, 0.2, 0.3.$$

For the active variables,

$$X_1, \dots, X_5 \sim N(0, 1),$$

and the mean signal is

$$\mu(X) = 0.30X_1 - 0.27X_2 + 0.24X_3 - 0.21X_4 + 0.18X_5.$$

We compare two scenarios.

3.1 Clean homoskedastic scenario

All inactive predictors are independent standard normal variables and

$$Y = \mu(X) + \varepsilon, \quad \varepsilon \sim N(0, 1).$$

This is a favorable baseline for Pearson SIS.

3.2 Shared-volatility scenario

Let

$$A_i = \exp(0.5Z_i), \quad Z_i \sim N(0, 1).$$

The active variables remain unscaled standard normal variables. The inactive variables and the noise component of the response share the same volatility:

$$X_{ij} = A_i E_{ij}, \quad j \notin S,$$

and

$$Y_i = \mu(X_i) + A_i \varepsilon_i.$$

Here E_{ij} , Z_i , and ε_i are independent standard normal variables.

Conditionally on the volatility factor, inactive predictors have no mean association with the response. However, the shared scale factor can generate large products $X_{ij}Y_i$ and can make ordinary correlation p-values anti-conservative.

4 How Points Are Drawn

For a fixed method and q level, each Monte Carlo replication produces a selected set $\hat{S}_q$. We compute

$$\text{FPR}_b(q) = \frac{|\hat{S}_q^{(b)} \cap S^c|}{|S^c|}$$

and

$$\text{SSP}_b(q) = 1\{S \subseteq \hat{S}_q^{(b)}\}.$$

The plotted point is

$$\left(\frac{1}{B} \sum_b \text{FPR}_b(q), \frac{1}{B} \sum_b \text{SSP}_b(q) \right).$$

The same selected set also gives a model-size measure:

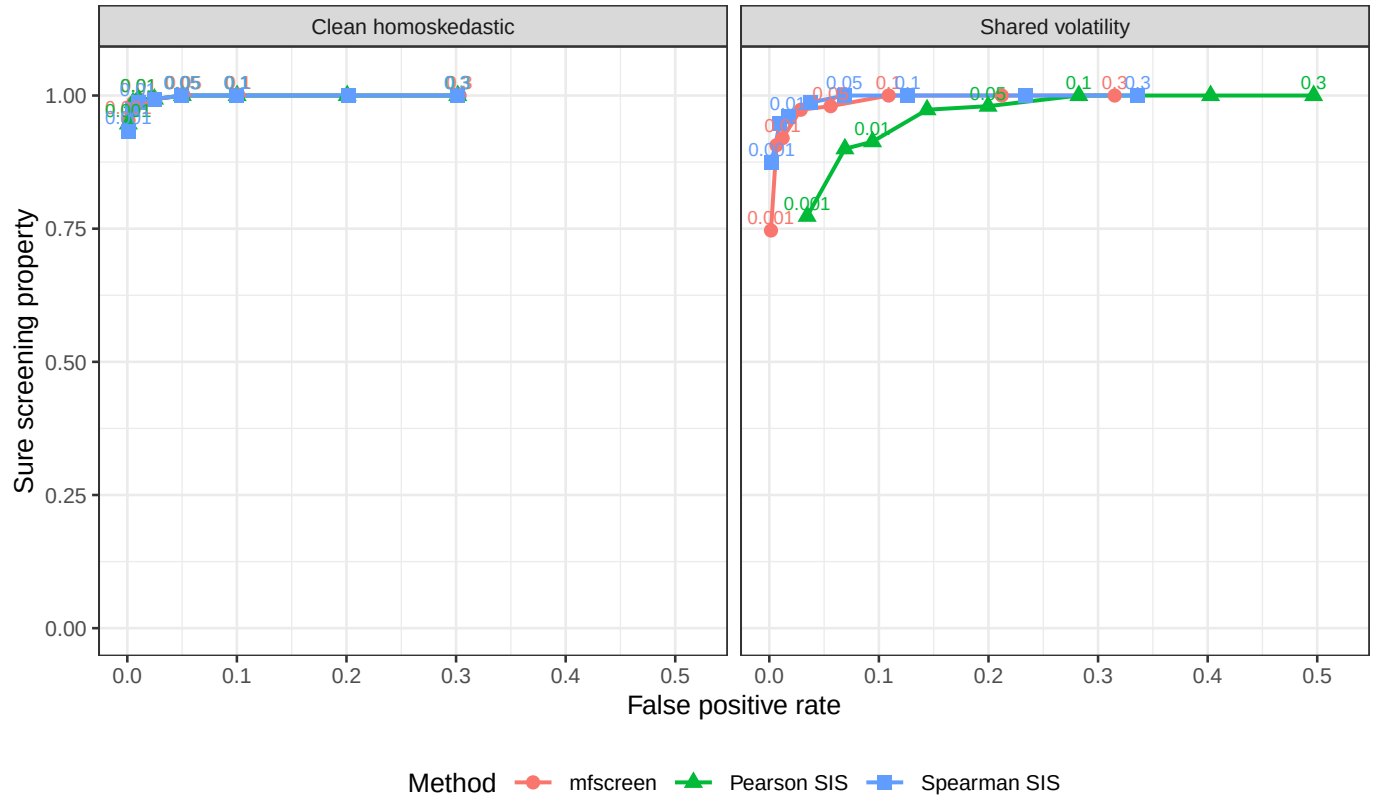
$$|\hat{S}_q|.$$

5 Results

5.1 Efficiency frontier

Efficiency frontier under shared volatility

$n = 1000$, $p = 1000$, active variables = 5, replications = 150

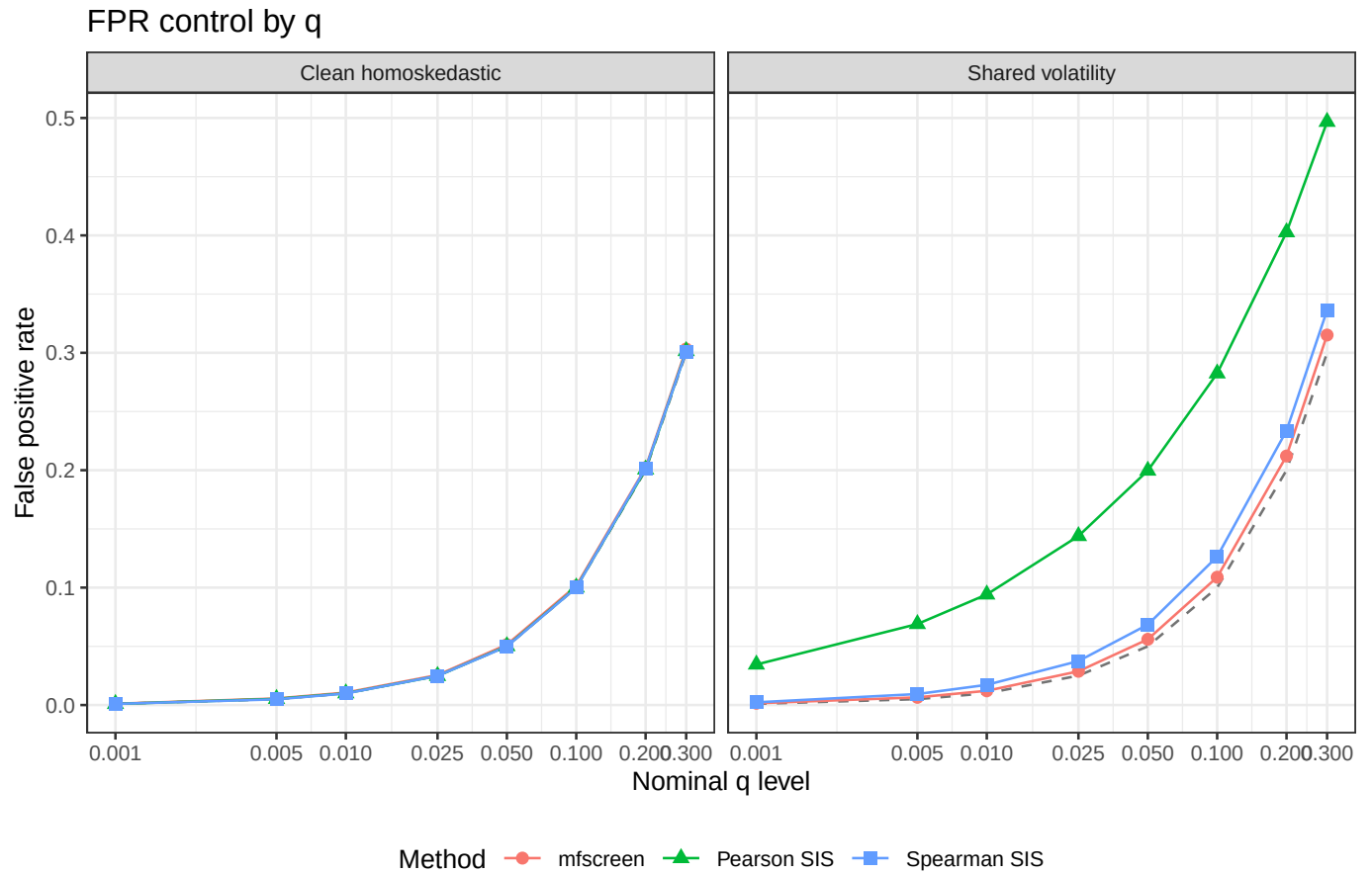


In the clean homoskedastic scenario, all three methods are similar. This is expected: the competitors are being used in a setting close to their ideal calibration.

The shared-volatility scenario is more revealing. Pearson SIS reaches high SSP, but does so with many false positives. For example, at $q = 0.05$, Pearson SIS has SSP about 0.98 but FPR about 0.20. In contrast, **mfscreen** has the same SSP about 0.98 but FPR about 0.056. Spearman SIS is more robust than Pearson, but still has FPR about 0.069 at $q = 0.05$, larger than **mfscreen**.

This is the efficiency advantage: for comparable sure screening probability, **mfscreen** produces a much smaller candidate set under shared volatility.

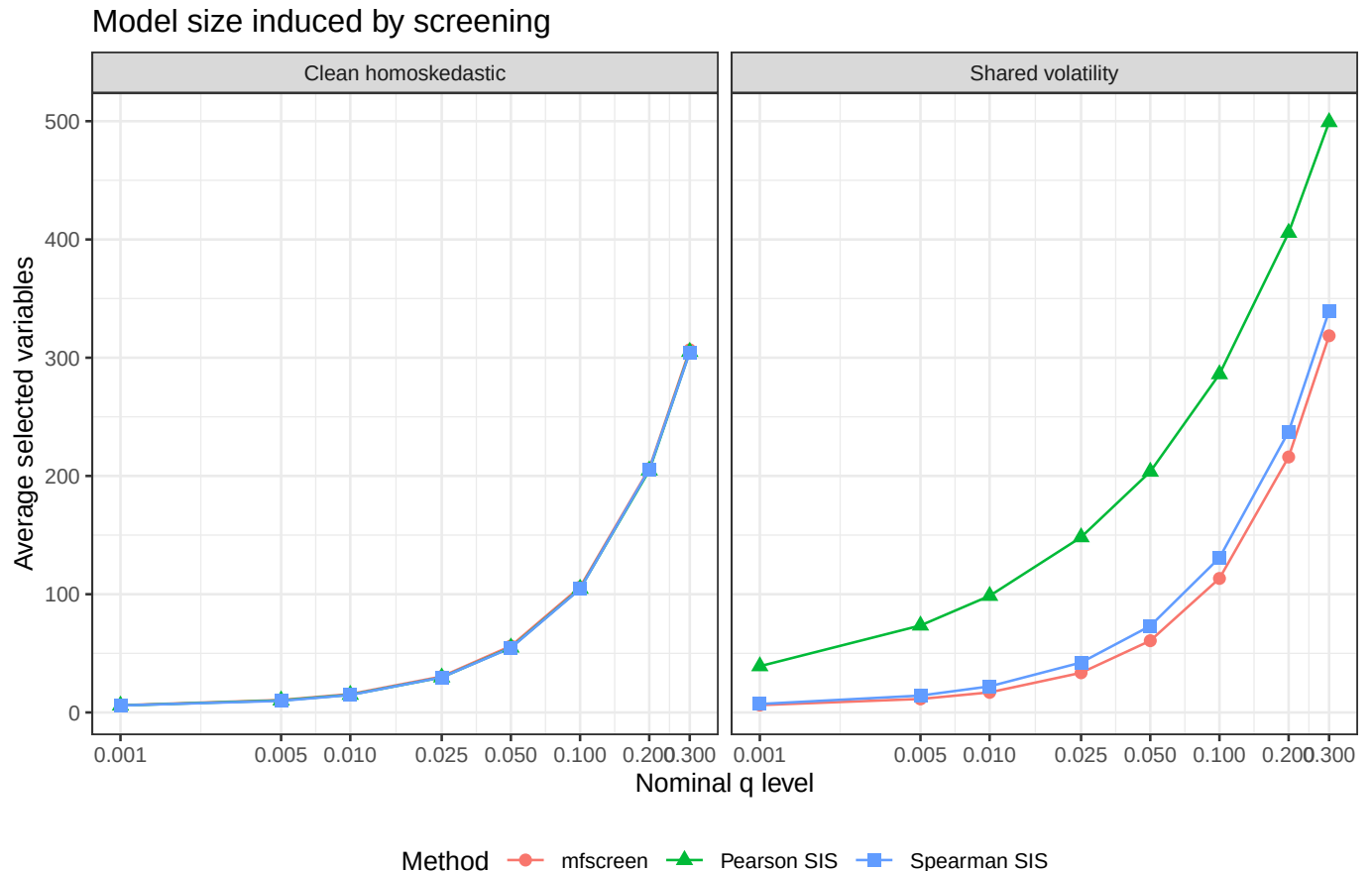
5.2 FPR calibration



The dashed line is the ideal calibration $FPR = q$. In the clean scenario, all methods follow this line closely. Under shared volatility, **mfscreen** remains close to the nominal q level, while Pearson SIS is strongly anti-conservative and Spearman SIS is moderately anti-conservative.

At $q = 0.01$, Pearson SIS has FPR about 0.094, roughly nine times the nominal level. At the same q level, **mfscreen** has FPR about 0.012. This difference is not a minor plotting artifact: it means Pearson includes many inactive variables before the downstream model-building step.

5.3 Number of selected variables



The model-size plot gives a practical interpretation of FPR inflation. In the shared-volatility scenario at $q = 0.05$, **mfscreen** selects about 61 variables on average, Spearman SIS selects about 73, and Pearson SIS selects about 204. All three methods have high SSP there, but Pearson pays for it by passing a much larger candidate set to later analysis.

6 Numerical Summary

Table 1: Mean FPR, SSP, and selected model size over Monte Carlo replications.

scenario	q	method	fpr	ssp	selected_count
Clean homoskedastic	0.001	mfscreen	0.0011	0.9533	6.0733
Clean homoskedastic	0.001	Pearson SIS	0.0009	0.9467	5.8867
Clean homoskedastic	0.001	Spearman SIS	0.0010	0.9333	5.9067
Clean homoskedastic	0.005	mfscreen	0.0056	0.9867	10.5333
Clean homoskedastic	0.005	Pearson SIS	0.0052	0.9867	10.2000
Clean homoskedastic	0.005	Spearman SIS	0.0049	0.9733	9.8133
Clean homoskedastic	0.010	mfscreen	0.0106	0.9933	15.5400
Clean homoskedastic	0.010	Pearson SIS	0.0100	0.9933	14.9533

scenario	q	method	fpr	ssp	selected_count
Clean homoskedastic	0.010	Spearman SIS	0.0098	0.9867	14.7800
Clean homoskedastic	0.025	mfscreen	0.0255	0.9933	30.3933
Clean homoskedastic	0.025	Pearson SIS	0.0248	0.9933	29.6333
Clean homoskedastic	0.025	Spearman SIS	0.0248	0.9933	29.6867
Clean homoskedastic	0.050	mfscreen	0.0515	1.0000	56.2067
Clean homoskedastic	0.050	Pearson SIS	0.0502	1.0000	54.9333
Clean homoskedastic	0.050	Spearman SIS	0.0499	1.0000	54.6200
Clean homoskedastic	0.100	mfscreen	0.1015	1.0000	106.0000
Clean homoskedastic	0.100	Pearson SIS	0.1002	1.0000	104.6733
Clean homoskedastic	0.100	Spearman SIS	0.0999	1.0000	104.4000
Clean homoskedastic	0.200	mfscreen	0.2023	1.0000	206.2600
Clean homoskedastic	0.200	Pearson SIS	0.2007	1.0000	204.6667
Clean homoskedastic	0.200	Spearman SIS	0.2016	1.0000	205.5867
Clean homoskedastic	0.300	mfscreen	0.3032	1.0000	306.7067
Clean homoskedastic	0.300	Pearson SIS	0.3015	1.0000	305.0000
Clean homoskedastic	0.300	Spearman SIS	0.3010	1.0000	304.4667
Shared volatility	0.001	mfscreen	0.0015	0.7467	6.2333
Shared volatility	0.001	Pearson SIS	0.0345	0.7733	39.1067
Shared volatility	0.001	Spearman SIS	0.0023	0.8733	7.1600
Shared volatility	0.005	mfscreen	0.0066	0.9067	11.4267
Shared volatility	0.005	Pearson SIS	0.0690	0.9000	73.5467
Shared volatility	0.005	Spearman SIS	0.0094	0.9467	14.2600
Shared volatility	0.010	mfscreen	0.0121	0.9200	16.9800
Shared volatility	0.010	Pearson SIS	0.0942	0.9133	98.6067
Shared volatility	0.010	Spearman SIS	0.0172	0.9600	22.0600
Shared volatility	0.025	mfscreen	0.0287	0.9733	33.5533
Shared volatility	0.025	Pearson SIS	0.1439	0.9733	148.1933
Shared volatility	0.025	Spearman SIS	0.0374	0.9867	42.2200
Shared volatility	0.050	mfscreen	0.0560	0.9800	60.6733
Shared volatility	0.050	Pearson SIS	0.1996	0.9800	203.6200
Shared volatility	0.050	Spearman SIS	0.0685	1.0000	73.1600
Shared volatility	0.100	mfscreen	0.1089	1.0000	113.3333
Shared volatility	0.100	Pearson SIS	0.2825	1.0000	286.0467
Shared volatility	0.100	Spearman SIS	0.1264	1.0000	130.7933
Shared volatility	0.200	mfscreen	0.2121	1.0000	216.0333
Shared volatility	0.200	Pearson SIS	0.4027	1.0000	405.6733
Shared volatility	0.200	Spearman SIS	0.2336	1.0000	237.4733
Shared volatility	0.300	mfscreen	0.3151	1.0000	318.5533
Shared volatility	0.300	Pearson SIS	0.4967	1.0000	499.2467
Shared volatility	0.300	Spearman SIS	0.3362	1.0000	339.5333

7 Conclusion

This simulation isolates a situation where ordinary correlation screening is not just less robust but practically inefficient. Shared volatility creates many false positives for Pearson SIS and, to a lesser extent, Spearman SIS. The model-free reciprocal statistic keeps the false positive rate close to the target while retaining the active variables with similar probability. This supports using **mfscreen** as a first-stage screen when high-dimensional data may contain scale heterogeneity, common volatility, or outlier-prone observations.